

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Chakradhar (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

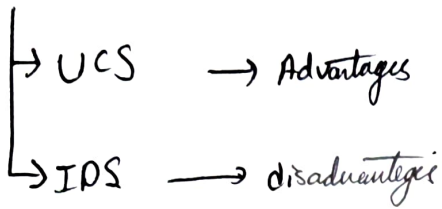
Chakradhar

1) A SMART home automation system

manag

- 1) lighting.
 - 2) Temperature
 - 3) security
- } Based on user behaviour.

2) MICE ROBO



15
20

1)

Smart HOME Automation.

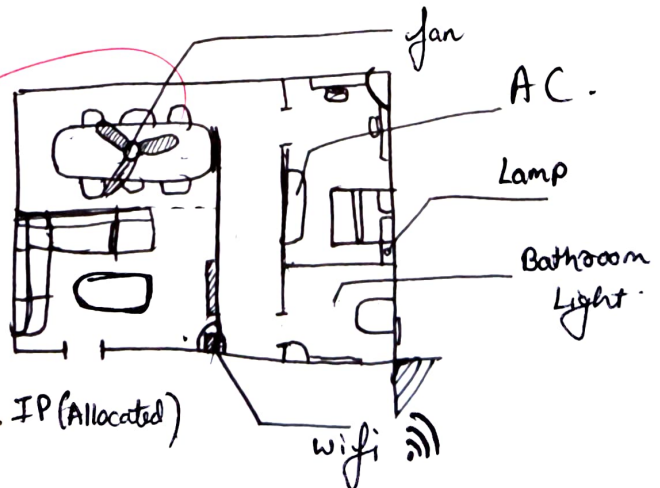
i) what is automation.

↳ making A task which is done by human solved is Automation (Repetitive tasks)

Examples

- ↳ vacuuming Robot
- ↳ Robotic Arm
- ETC.

Home automation using ESP-32 Dev Board
And local multimode network



1) every Device connected to the wifi.

2) when device connected to wifi. IP (Allocated)

3) Base server [Master]

4) Devices [Slaves]

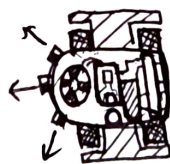
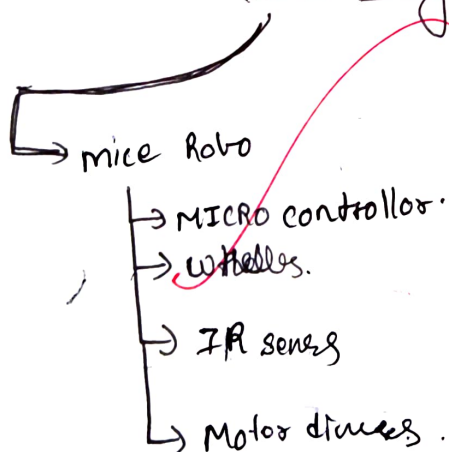
- 1) This system works on the Master slave concept.
- 2) The Master provides Action/Information from slaves.
- 3) Master stores and uses the information for future use.
- 4) All commands are performed by the slaves through IPs.
- 5) 2 ways of using USER Data.

1) Regular uses Time at particular part of hour.

2) Auto mated. Receives sensor command for & back up

6 This type of system comes under IOT based system use cases.

Maze Solving Robo.



2 ways AMICE solves Mazes

1) UCS

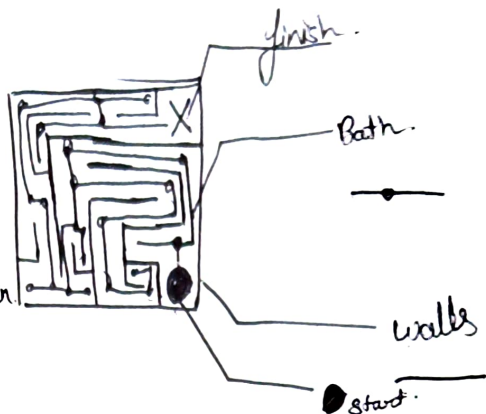
2) IOS

VCS $\rightarrow$ Searches the full maze uniformly and finds the most optimal solution.

Advantages

$\rightarrow$ Reliable (Very Reliable)

$\rightarrow$ Efficient during solve Run.



Con's

$\rightarrow$ Slow during Mapping Run.

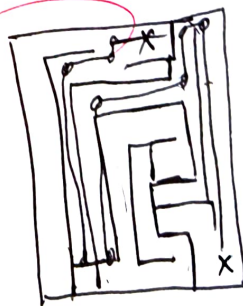
$\rightarrow$ Too much memory usage.

2) IDS $\rightarrow$ Take the most shortest path and give output ~~rather~~ for faster results.

Advantages

$\rightarrow$ fast during searching Run.

$\rightarrow$ Time efficient



Con's

$\rightarrow$ not the best suit.

$\rightarrow$ might not get best score less time & energy Time.